
Envisage: Diffusion-Based Rhinoplasty Goal Visualization with Mask-Decomposed Evaluation

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Abstract

Localized generative editing needs localized evaluation: full-image identity metrics are structurally confounded under hard-composited edits. We present **Envisage**, a FLUX.1-Fill inpainting reference pipeline for rhinoplasty goal visualization from a single frontal photograph. The pipeline combines 24 landmark-indexed clinical presets, MediaPipe masks, and hard-mask compositing. The composite preserves outside-mask pixels by construction, so full-face identity scores are dominated by copied pixels rather than by the diffusion backbone. We therefore introduce **SurgicalScore**, a mask-decomposed 0–1 protocol scoring edit direction, edit magnitude, masked LPIPS, realism, and outside-mask SSIM; its raw composite assigns 0.919 [0.918, 0.920] to a perfect-predictor control, anchoring the ceiling.

On $N=211$, the paired ArcFace gain (output-to-GT minus input-to-GT) is negative for all methods (Envisage -0.048 smallest, vs. ICEdit -0.139 , Kontext -0.242 , InstructPix2Pix -0.294 ; $p < 10^{-4}$), with external validation on a 457-pair ASPS/PCA corpus showing the same negative gap. This decrease is due to the in-mask edit failing to shift identity toward post-op: copied outside-mask pixels are identical between input and output, so the paired difference is dominated by the in-mask shift. With SurgicalScore, Envisage achieves the highest score (0.599 [0.579, 0.619]); the divergence from the ArcFace ranking suggests global identity is the wrong proxy for localized surgical accuracy. Furthermore, a 5-seed oracle reduces the residual ArcFace gap by 73% (-0.054 to -0.015), with positive output-to-GT gain on 33.9% of cases; SurgicalScore is also raised from 0.609 to 0.743 [0.725, 0.762] ($N=207$), indicating candidate-space headroom for a learned ranker. As such, the path forward for surgical-prediction runs through metrics that measure the edit, not the face around it. We release Envisage, SurgicalScore, preset definitions, and matched split manifests.

1 Introduction

Rhinoplasty accounts for approximately 180,000 procedures annually in the United States [American Society of Plastic Surgeons, 2024b]. At this volume, expectation management is the dominant determinant of patient satisfaction; revision rates of 5–15% [Neaman et al., 2013, Newman et al., 2024] reflect, in part, discrepancies between expected and actual outcomes. Current visualization tools span surgeon-drawn sketches and 2D overlays at one end to proprietary 3D simulation systems such as Crisalix and Vectra 3D at the other. The 3D systems require capital investment in stereophotogrammetry hardware and dedicated capture space, plus per-clinic licensing and trained operators; these costs restrict access to high-volume metropolitan practices and exclude solo, rural,

and community settings. Sketches and 2D overlays, on the other hand, are surgeon-dependent, single-view, and not photorealistic. The issue is clear: we need a tool that is believable, affordable, and faithful to the surgical scope. That means photorealistic outputs, no specialized capture hardware, and architectural preservation of non-surgical regions. No existing system delivers all three.

The gap has two structural roots. First, the methods themselves fall short: geometric methods [Bookstein, 1989] cannot synthesize new skin texture, 3D approaches require CT scans or meshes [Ma et al., 2021] unavailable in standard clinics, and full-face diffusion models drift identity across non-surgical regions. Second, the metric is broken too: full-face evaluation conflates two distinct objectives (outside-mask identity preservation and inside-mask surgical accuracy) and reports a single number that mixes them.

LandmarkDiff [Anonymous, 2025] established the failure mode. That system conditioned SD 1.5 on landmark wireframes and regenerated the entire face before compositing the surgical region back. Decomposed evaluation showed over 95% of its identity score came from copied pixels: 0.509 composited, 0.023 without. The result exposed five architectural limitations: low-resolution backbone, sparse conditioning, full-face regeneration, synthetic training data, post-hoc compositing. Envisage corrects all five.

We invert the formulation. Rather than regenerate the full face and composite the surgical region back, the pipeline masks the surgical region from the outset, modifies a monocular depth map at landmark-indexed Gaussians to encode the target tissue change, runs a pretrained depth-conditioned diffusion model on the mask only, and composites the output verbatim with the input outside the mask. Outside-mask identity preservation is an architectural guarantee, not a backbone assumption: the only quantity left to measure is inside-mask edit fidelity against paired postoperative ground truth.

That formulation produces the smallest GT-vs-BL ArcFace gap of any method evaluated on an $N=211$ ASPS+PCA cohort (Envisage -0.048 vs. ICEdit -0.139 , Kontext -0.242 , IP2P -0.294 ; $p < 10^{-4}$ paired) and the highest mask-decomposed SurgicalScore (0.599 [0.579, 0.619] vs. 0.502 / 0.337 / 0.229). The result replicates on a 457-pair external corpus. No method, including ours, achieves positive paired ArcFace gain over the unedited input proxy: this is a finding about the metric, not a failure to report. ArcFace was trained for full-face verification across changes orders of magnitude larger than rhinoplasty; it cannot grade an edit at 5%-of-pixels scale.

Our contributions.

1. **Methodological diagnosis.** Full-image identity metrics are structurally confounded under hard-composited edits. Outside-mask SSIM exceeds 0.999 for FLUX backbones wrapped in our composite (Appendix J); the property is rule-driven, not backbone-driven, and generalizes beyond rhinoplasty (Appendix K).
2. **SurgicalScore.** A mask-decomposed 0–1 protocol scoring landmark edit direction, magnitude, masked LPIPS, realism, and outside-mask preservation. Released with matched HDA paired pre/post splits ($N=27$ bleph, $N=21$ rhino, $N=9$ rhytid). The raw composite assigns 0.919 [0.918, 0.920] to a perfect-predictor control, anchoring the ceiling.
3. **Cross-method, cross-procedure characterization.** No method exceeds the input proxy under GT ArcFace on $N=211$. Envisage produces the only positive-gap fraction above 5% (16.1% vs. 0.0 / 4.3 / 1.0%). SurgicalScore reveals component-level failure modes that ArcFace alone hides.
4. **Envisage reference pipeline.** Depth-conditioned FLUX-Fill inpainting with 8 rhinoplasty presets (Daniel’s taxonomy) plus 8 blepharoplasty (Tessier) and 8 rhytidectomy (SMAS) preset definitions released as infrastructure. Smallest GT-vs-BL ArcFace gap and highest SurgicalScore of any method evaluated.

Component-level evidence. On $N=211$ ($B=10,000$, $K=5$ best-of paired permutation), each component is individually paired-significant: hard-mask composite $\Delta SS = +0.034$ ($p=0.001$), 24-preset depth modification $\Delta SS = +0.034$ ($p=0.002$), depth-ControlNet conditioning $\Delta SS = +0.023$

($p=0.030$, Appendix P). Equipping baselines with the same composite closes most of the bare-baseline gap; Envisage still leads ICEdit+composite by $+0.058$ ($p < 10^{-3}$) and IP2P+composite by $+0.045$ ($p=0.008$, Appendix R). K= 5 best-of upper-bounds the deployed-system contribution; recovering the gain requires a learned ranker (Appendix A: naive rankers do not).

2 Related Work

Surgical prediction. Eldaly et al. [2022] survey simulation and AI methods for rhinoplasty; the surveyed geometric warp approaches cannot synthesize new skin texture. Knoedler et al. [2024] train a pix2pix GAN on lateral-profile silhouettes ($N=3,030$ pairs) and report 52.5% Visual Turing Test discrimination. Frontal and internal views are excluded by dataset construction, leaving the frontal regime open. Bini et al. [2025] apply inpainting and depth estimation to facial reconstruction, a different task from aesthetic rhinoplasty on intact anatomy. Loo-Duque et al. [2024] apply diffusion to septorhinoplasty visualization but evaluate without paired ground-truth or identity metrics. Varghaei et al. [2025a] report landmark-based aesthetic outcome assessment on 1,259 paired patients ($N=366$ rhinoplasty), establishing the geometric-plus-identity evaluation regime our SurgicalScore extends. PtoisDiffusion [Huang et al., 2024] uses training-free ControlNet guidance for blepharoptosis, a precedent for zero-shot surgical simulation; we extend it to depth-conditioned blepharoplasty, rhinoplasty, and rhytidectomy. 3D methods [Ma et al., 2021] require volumetric or multi-view input, unavailable in standard clinics.

Diffusion models and ControlNet. Latent diffusion models [Rombach et al., 2022] generate photorealistic images from text and spatial conditioning; ControlNet [Zhang et al., 2023] adds spatial control to frozen backbones via zero-convolution layers. Depth conditioning maps tissue displacement directly to surface change, a connection edge or pose conditioning does not provide. ICEdit [Zhang et al., 2025] applies instruction-based editing via FLUX.1-Fill-dev with a mixture-of-experts LoRA and a diptych formulation; we evaluate it as a baseline (Section 5).

Identity preservation. ArcFace [Deng et al., 2019] cosine similarity is the standard identity metric in face research. It conflates identity change in the surgical region (expected) with identity drift in non-surgical regions (undesired); our mask-decomposed evaluation separates these terms. Face-recognition fairness on darker skin tones is documented in the broader literature [Buolamwini and Gebre, 2018], but no prior surgical-prediction work reports metrics stratified by the Monk Skin Tone Scale [Monk, 2023].

3 Methods

3.1 Pipeline Overview

We invert the formulation: Envisage masks the surgical region from the outset rather than regenerating the full face and compositing the surgical region back. A procedure-specific text prompt provides the semantic direction of change; a modified monocular depth map encodes the intended surface displacement. After denoising, a hard-mask composite copies all non-surgical pixels verbatim from the input.

Outside-mask identity preservation is an architectural guarantee, not a backbone assumption. The remaining modeling problem is inside the mask: edit fidelity against paired postoperative ground truth.

The three pipeline stages are: (1) *TPS pre-warp* for procedure-specific geometric deformation; (2) *depth modification* to encode profile changes on the monocular depth map; and (3) *FLUX inpainting* to regenerate the masked region, conditioned on the modified depth map via a pretrained ControlNet.

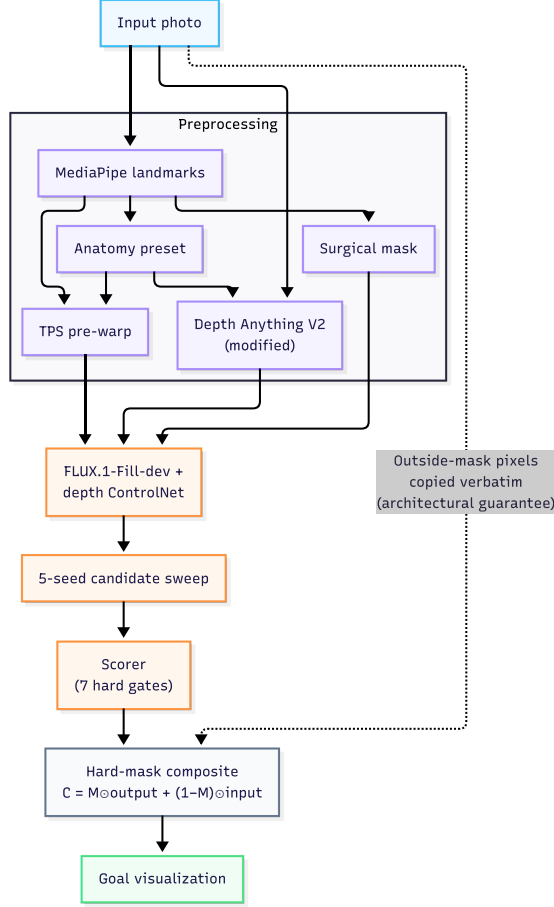


Figure 1: **Envisage pipeline overview.** A preoperative photograph is processed in three stages: (1) a procedure-specific TPS pre-warp displaces nasal landmarks by 2–4 px; (2) Depth Anything V2 estimates a monocular depth map, which is then modified by landmark-indexed Gaussian kernels to encode the intended tissue displacement; and (3) FLUX.1-Fill-dev, conditioned on the modified depth map via a pretrained depth ControlNet, regenerates only the masked surgical region across a 5-seed candidate sweep. A 7-gate scorer selects the winning candidate, and a hard-mask composite pastes all non-surgical pixels verbatim from the input, so outside-mask identity preservation is architectural rather than a backbone property. We use the *goal visualization* framing throughout, not patient-specific outcome prediction.

128 3.2 Mask Generation

129 For rhinoplasty, 30 MediaPipe FaceLandmarker landmarks (478-point successor of the 468-vertex
130 face mesh [Kartynnik et al., 2019]) spanning the nasal dorsum, tip, and alar wings define the convex-
131 hull mask, dilated 25 pixels with an elliptical structuring element and Gaussian-feathered ($\sigma=15$
132 pixels). The preset system (Section 3.5) supports 8 frontal-detectable rhinoplasty sub-procedures
133 whose masks extend the nasal core to include the alar base, bridge sidewalls, and supratip region as
134 needed. The blepharoplasty mask spans 44 upper-lid and brow landmarks; the rhytidectomy mask
135 spans 36 jaw-contour landmarks. Both follow the same dilation and Gaussian-feathering protocol as
136 the rhinoplasty mask. Their preset definitions are released as part of the framework but are not the
137 primary evaluation target of this paper.

138 3.3 TPS Pre-Warp

139 For rhinoplasty, bridge sidewall landmarks (indices 193, 245, 188, 174, 217 and their right-side
140 mirrors) are displaced inward by 3–4 pixels, tip lobule landmarks inward by 2–3 pixels, and alar
141 landmarks fixed at zero displacement. The TPS uses `scipy.interpolate.RBFInterpolator`
142 with a thin-plate spline kernel and boundary anchor points to prevent deformation outside the surgical
143 region.

144 3.4 Depth Modification

145 Monocular depth was estimated with Depth Anything V2 Small [Yang et al., 2024], producing
146 $D \in \mathbb{R}^{H \times W}$ normalized to $[0, 255]$. Modified depth is:

$$D'(x, y) = M(x, y) \cdot [D(x, y) - \sum_k \alpha_k \cdot G_k(x, y; c_k, \sigma_k)] + [1 - M(x, y)] \cdot D(x, y), \quad (1)$$

147 where M denotes the binarized surgical mask used for depth modification (the feathered image-
148 composite mask is a separate, image-domain quantity), each G_k is a 2D Gaussian at a landmark, and
149 α_k controls displacement magnitude. For rhinoplasty, two Gaussians are applied: a broad one at
150 the nasion (landmark 6, $\sigma_x=0.05w$, $\sigma_y=0.10h$, $\alpha=100$) for dorsal hump reduction, and a tight one
151 at the nasal tip (landmark 1, $\sigma_x=0.015w$, $\sigma_y=0.15h$, $\alpha=115$) for the supratip break. Amplitudes
152 α_k are landmark-indexed hand-tuned constants calibrated once on a held-out HDA case and held
153 fixed for all reported evaluation runs; evaluation uses 50% default scaling, applied uniformly without
154 per-cohort retuning. Depth modification does not distinguish bony from cartilaginous deformation or
155 model surgeon-specific choices such as spreader-graft placement [Daniel, 2010].

156 3.5 Anatomy-Aware Preset System

157 We encoded 8 of the 75 atomic rhinoplasty sub-procedures from Daniel’s *Mastering Rhinoplasty*
158 taxonomy [Daniel, 2010] that are detectable from a single frontal photograph: dorsal hump reduction,
159 tip narrowing, bridge narrowing, alar base narrowing, dorsal straightening, tip definition, tip rotation
160 up, and nose shortening. Each sub-procedure has a landmark-measurement detection threshold and a
161 prompt fragment. The prompt builder selects the top 3 detected sub-procedures by a fixed clinical-
162 priority ordering (PRIORITY_ORDER in `rhino_config.py`, derived from Daniel’s taxonomy
163 frequency rankings) and concatenates their fragments, producing a subject-specific prompt. The
164 blepharoplasty (8) and rhytidectomy (8) preset definitions follow the same architecture using Tessier
165 orbital and SMAS-layer anatomy respectively; they are released as part of the framework.

166 3.6 Diffusion Inpainting

167 FLUX.1-Fill-dev [Black Forest Labs, 2024], a 12-billion-parameter rectified flow
168 fill model, is used with a pretrained depth ControlNet [Zhang et al., 2023]
169 (`jasperai/Flux.1-dev-Controlnet-Depth`) at conditioning scale 0.5. The
170 FluxInpaintPipeline uses guidance scale 3.5, 20 denoising steps, BF16 precision, and
171 512×512 resolution. Inpainting strength for rhinoplasty is 0.75 at 50% depth modification intensity.
172 No procedure-specific training is required for the base pipeline (a separate DPO LoRA probe is
173 reported in Appendix H).

174 3.7 Mask-Decomposed Evaluation: SurgicalScore

175 Standard full-face metrics conflate two distinct objectives: outside-mask fidelity to the input, and
176 inside-mask accuracy against the ground-truth postoperative image. We define **SurgicalScore**, which
177 decomposes evaluation by mask region and reference:

- 178 • **Outside-mask SSIM**: prediction vs. input on pixels where $M < 0.5$. Approaches 1.0 for
179 methods preserving non-surgical regions.
- 180 • **Inside-mask LPIPS**: prediction vs. ground truth on the mask bounding crop (AlexNet
181 backbone [Zhang et al., 2018]).

- **Input ArcFace:** output vs. input cosine similarity (identity preservation).
- **GT ArcFace:** output vs. ground truth cosine similarity (postoperative target-proximity proxy in identity space).
- **BL ArcFace:** input vs. ground truth (dataset-level baseline, not a method score).

Because the composite $C = M \odot G + (1 - M) \odot I$ differs from the input only on the masked region, the operational identity preservation reduces to the directly measurable outside-mask SSIM, which exceeds 0.999 across all evaluated cohorts (a property of the compositing rule, not of any backbone). Empirically, the local Lipschitz constant of ArcFace under masked perturbations is bounded by $L_{p95} \leq 1.2 \times 10^{-5}$ ($N=255$ samples, Appendix M).

A **Mask-Area Modifier** corrects small-mask inflation: metrics are computed on a resize-to-256 crop of the mask bounding box to normalize the effective contribution of the surgical region. The affine rescale (slope 0.70, intercept 0.30) anchors the passthrough floor at 0.30, so any SurgicalScore below 0.30 is scored below returning the input under the protocol. Component weights are protocol hyperparameters; the floor is set by the affine rescale, not by the weights.

This decomposition prevents the confound observed in prior work [Anonymous, 2025], where over 95% of the identity score originated from composited pixels (0.509 composited vs. 0.023 without).

SurgicalScore composite. For scored validation cases, the five components are aggregated as:

$$SS = 0.30 + 0.70 \cdot \frac{R_O - R_I}{1 - R_I}, \quad R_O = 0.40 A + 0.30 B + 0.15 C + 0.10 D + 0.05 E \quad (2)$$

where R_I is R_O evaluated with $O = I$ (the passthrough reference), and a hard ArcFace identity gate disqualifies candidates when $\cos(I, O) < 0.65$ (Input ArcFace, input-to-output). Component weights balance directional and fidelity terms; the passthrough floor at 0.30 is set by the affine rescale (slope 0.70, intercept 0.30). Both are protocol hyperparameters fixed before any cross-method comparison. SurgicalScore is the diagnostic protocol; cross-method ranking uses the GT-vs-BL ArcFace gap, an independent embedding metric. The five components: A = directional alignment (cosine of morphometry edit vectors, as in Gal et al. [2022]); B = edit magnitude fit (asymmetric log-ratio, over-edit penalized at $1.5\times$, under-edit at $1.0\times$); C = masked LPIPS fidelity on a 256×256 mask-crop resize (Mask-Area Modifier applied); D = face realism: mean of SER-FIQ [Terh r st et al., 2020] and CR-FIQA [Boutros et al., 2023] quality scores, with a three-signal proxy (embedding norm, Laplacian variance, color balance) as fallback when FIQA models are unavailable; E = outside-mask soft preservation ($1 - \text{LPIPS}_{\text{out}}/\tau$, $\tau=0.10$). Gal et al. [2022] introduced directional cosine loss in image editing; A adapts this to procedure morphometry space. The rhinoplasty morphometry vector $\phi_{\text{morph}}^{\text{rhino}} \in \mathbb{R}^5$ contains: Goode ratio [Goode, 1984] (lateral-view approximation computed from frontal photographs: tip projection / nasal length); nasolabial angle (lateral-view approximation computed from frontal photographs, subnasale, landmark 2); alar width / intercanthal distance; dorsal line RMS deviation / nasal length; nasal length / face height. All components are scale-invariant ratios or normalized angles.

SS_{raw} calibration anchor. SurgicalScore is the calibrated metric used for all cross-method comparison. The uncalibrated raw composite $R_O = 0.40 A + 0.30 B + 0.15 C + 0.10 D + 0.05 E$ (without the passthrough-anchored affine rescale) is reported alongside as a calibration check; we refer to it as SS_{raw}. Under SS_{raw}, an oracle control in which the system output is replaced directly by the ground-truth postoperative image (no hard-mask composite) achieves 0.919 [0.918, 0.920] on the $N=211$ cohort ($B=10,000$ bootstrap). This confirms the metric ceiling is achievable for a perfect predictor and that the score range $[0, 1]$ is used as intended. Aggregate Envisage SS_{raw} on $N=211$ is 0.563 [0.536, 0.590]. SS_{raw} values for all ablation variants appear in Table 19.

Component-level discrimination. Each of the five SurgicalScore components flags failures the others miss. Case Nose_33 receives outside-mask preservation score (E) = 0.000 from a compositing artifact in the mask boundary region despite high directional alignment ($A = 0.997$) and edit

228 magnitude ($B = 0.872$), traced to `evaluation/surgicalscores_v5_fixed_hda_rhino.json`.
 229 Case Nose_154 receives directional alignment ($A = 0.004$) from a preset selection mismatch despite
 230 intact realism ($D = 0.738$) and outside-mask preservation ($E = 0.540$). Case Nose_161 receives
 231 edit magnitude ($B = 0.049$) from insufficient in-mask change despite high directional alignment
 232 ($A = 1.000$) and realism ($D = 0.736$). These three cases are not discriminable on full-face ArcFace
 233 alone: their $\cos(\text{input}, \text{output})$ scores are 0.740, 0.858, and 0.692, respectively, all above the identity
 234 gate threshold.

235 3.8 Decision Model

236 Seven hard gates must be passed for a candidate to be valid: (1) identity: Input ArcFace ≥ 0.65 ; (2)
 237 outside-SSIM: ≥ 0.95 ; (3) landmark drift: ≤ 15.0 px; (4) dark-hole: mask HSV-V < 20 area $\leq 0.5\%$;
 238 (5) color consistency: mean hue shift $\leq 15.0^\circ$; (6) crease: blepharoplasty only; (7) procedure fidelity:
 239 landmark-atlas severity bands. Passing candidates are ranked by a weighted composite (0.25 identity,
 240 0.15 outside-SSIM, 0.15 landmark drift, 0.15 color consistency, 0.15 cross-method agreement, 0.10
 241 TPS agreement, 0.05 aesthetic). Cross-method agreement is computed only among Envisage’s own
 242 candidate seeds and the deterministic TPS/preset proxies; no baseline outputs or postoperative ground
 243 truth are used for candidate selection. If no diffusion candidate passes, the deterministic TPS warp
 244 ($M5$) is substituted as a non-generative geometric fallback.

245 4 Experiments

246 4.1 Dataset

247 We use two cohorts: HDA [Rathgeb et al., 2020] for matched paired pre/post photographs across
 248 procedures, and an external ASPS+PCA rhinoplasty cohort for the headline comparison. The
 249 HDA Plastic Surgery Database contains 638 subjects. The matched HDA test splits contain 27
 250 blepharoplasty, 21 rhinoplasty, and 9 rhytidectomy pairs. The headline rhinoplasty comparison
 251 (Section 5, Table 1) uses the $N=211$ ASPS/PCA cohort described below. The matched 21-pair HDA
 252 rhinoplasty split is used for the DPO pilot probe (Appendix H) and for prior-work comparability in
 253 Appendix F.

254 We further assembled an external validation corpus from ASPS Photo Gallery [American Society of
 255 Plastic Surgeons, 2024a] and an additional private clinical archive of paired pre/post photographs
 256 (PCA), filtered to frontal views (InsightFace yaw-symmetry ≥ 0.82) and pre/post identity floor
 257 (≥ 0.35 ArcFace cosine), yielding 457 rhinoplasty pairs. All external pairs are held out from training,
 258 hyperparameter selection, and preset design.

259 HDA is the only public corpus with identity-paired pre/post photographs across the three pro-
 260 cedures; CelebA-HQ [Karras et al., 2018] and FFHQ [Karras et al., 2019] contain none, com-
 261 mercial 3D platforms are proprietary, and SurFace1259 [Varghaei et al., 2025b] was unavailable.
 262 The identity-floor filter excludes pairs whose appearance change exceeds plausible surgical drift;
 263 rhytidectomy_Facelift_08 is removed after manual review identified two different subjects.

264 4.2 Baselines

265 Three instruction-conditioned baselines were evaluated under two protocols: the $N=211$ ASPS+PCA
 266 cohort (used for the headline comparison in Table 1) and the matched $N=21$ HDA paired pool (used
 267 for prior-work comparability in Appendix F): (1) **InstructPix2Pix** [Brooks et al., 2023], a full-image
 268 text-instruction editor applied with the same preset-derived prompts; (2) **ICEdit** [Zhang et al., 2025],
 269 FLUX.1-Fill-dev with a MoE-LoRA via diptych inference; (3) **FLUX.1-Kontext-dev** [Batifol et al.,
 270 2025], a text-conditioned image-to-image variant run zero-shot followed by the same hard-mask
 271 composite used in Envisage.

272 Three internal diagnostic baselines accompany the main comparators: Direct Copy (Input ArcFace
 273 $= 1.0$ upper bound), TPS Warp (geometric morphing only), and FLUX Inpainting without ControlNet.

4.3 Implementation

Inference runs on a single NVIDIA L40S GPU (48 GB) in BF16. Depth estimation uses Depth Anything V2 Small [Yang et al., 2024]. Landmarks use MediaPipe FaceLandmarker (478 points; successor of the 468-vertex face mesh [Kartynnik et al., 2019]). ArcFace uses InsightFace buffalo_1 (R50). LPIPS uses AlexNet [Zhang et al., 2018]. Inference takes approximately 20 seconds per image. Training details for the DPO LoRA are given in Appendix H.

4.4 Statistical Inference

We computed 95% confidence intervals (CIs) on reported means wherever per-case or seed-level replicates were available, via percentile bootstrap ($B=10,000$ resamples, seed 42). For Table 4 Envisage rows, matched- N per-case GT Arc scores were not available locally; CIs were bootstrapped over five seed-level means (denoted †). ICEdit CIs were bootstrapped over per-case scores ($N=21$). IP2P and Kontext per-case data were not available; their CIs are not reported.

Bootstrapping over seed-level means captures inference variance (sensitivity to random initialization) but not case-level variance (sensitivity to which 21 cases constitute the test set). The reported 95% CIs are therefore not estimates of case-sampling uncertainty: they reflect seed-to-seed reproducibility, not the full uncertainty that would be estimated from a larger rhinoplasty population.

Within-protocol Input ArcFace ordering: Envisage exceeded ICEdit on rhinoplasty ($p=0.0361$, $N_{\text{ours}}=16$, $N_{\text{ice}}=21$, two-sample 10,000-permutation, two-sided). This test is included for protocol consistency; cross-method ordering on Input ArcFace is not informative of inside-mask quality, since the metric is structurally confounded by hard-mask compositing. For the $N=211$ cohort, paired sign-flip permutation tests against each baseline are reported in Table 2 ($p < 10^{-4}$ all three). For the matched $N=21$ DPO probe, per-case zero-shot GT ArcFace was not stored at evaluation time, so the probe is directional only.

5 Results

Cohorts. External corpus assembly: 457 raw rhinoplasty pairs from ASPS Photo Gallery ($N=441$) and PCA ($N=16$). Two filtered cohorts derive from this pool: (a) the $N=211$ cohort (ASPS = 202, PCA = 9) requires both MediaPipe landmark detection and pre/post ArcFace ≥ 0.65 ; this is the headline cohort. (b) The gate-pass $N=438$ subset (ASPS = 422, PCA = 16) requires only ArcFace ≥ 0.65 on at least one of five seeds; reported as the external gate-pass rate in Section 5.1 and the source-stratified pooled GT-vs-BL replication in Appendix I. (c) Matched $N=21$ HDA rhinoplasty pool, used for the DPO probe and Appendix F.

5.1 Headline ArcFace and SurgicalScore on Rhinoplasty

ArcFace headline. Table 1 gives the headline result on the $N=211$ rhinoplasty cohort: ASPS public gallery ($N=202$) and a private clinical archive (PCA, $N=9$), filtered to cases passing MediaPipe landmark detection and pre/post same-person ArcFace at 0.65 cosine. Across all four methods evaluated, no method achieves positive mean GT ArcFace gain over the unedited preoperative input. Envisage achieves output-to-GT cosine 0.662 versus input-to-GT 0.711, gap -0.048 (95% CI $[-0.055, -0.042]$, $B=10,000$); ICEdit, Kontext, and IP2P gaps are -0.139 , -0.242 , -0.294 with non-overlapping CIs (Table 1). On the four-way detected intersection ($N=205$), Envisage reduces the gap by $+0.090$ vs. ICEdit, $+0.189$ vs. Kontext, and $+0.245$ vs. IP2P, all paired sign-flip permutation $p < 10^{-4}$ (Table 2). The fraction of cases where output exceeds input proxy under GT ArcFace is 16.1% for Envisage versus 0.0/4.3/1.0% for the baselines.

SurgicalScore. Mean SurgicalScore on Envisage on this cohort is 0.599 [0.579, 0.619] vs. ICEdit 0.502 [0.467, 0.535], InstructPix2Pix 0.337 [0.295, 0.380], and FLUX.1-Kontext-dev 0.229 [0.188, 0.271] (Appendix N, Table 14). On the four-way detected intersection ($N=188$), Envisage

Table 1: **Main result on the $N=211$ rhinoplasty cohort** (ASPS public gallery $N=202$ + PCA private clinical archive $N=9$, passing MediaPipe landmark detection and same-person ArcFace at 0.65 cosine). **No method achieves positive mean GT ArcFace gain over the unedited preoperative input.** Envisage achieves the smallest gap of any method evaluated, with 16.1% of cases positive (0.0/4.3/1.0% for ICEdit/Kontext/IP2P). Per-method N varies due to InsightFace detection failures on a subset of generated outputs; BL Arc is computed on the corresponding per-method input subset, so each method’s gap is paired within its own attrition mask. Gap 95% CIs are percentile bootstrap over per-case scores ($B=10,000$, seed 42). % Pos. is the fraction of cases where output-to-GT ArcFace exceeds input-to-GT ArcFace; Envisage is the only method exceeding 5%. Paired statistical comparisons against Envisage are reported in Table 2.

Method	N	GT Arc $\uparrow$	BL Arc	Gap [95% CI]	% Pos. $\uparrow$
InstructPix2Pix	208	0.417	0.711	-0.294 [-0.331, -0.256]	1.0
ICEdit	206	0.573	0.712	-0.139 [-0.148, -0.130]	0.0
FLUX.1-Kontext-dev	211	0.469	0.711	-0.242 [-0.258, -0.225]	4.3
Envisage (ours)	211	0.662	0.711	-0.048 [-0.055, -0.042]	16.1

Table 2: **Paired comparison against Envisage on the four-way detected intersection ($N=205$).** Each baseline is restricted to the cases where all four methods produced a detected face, and the per-case ArcFace gap (output-to-GT minus input-to-GT) is differenced row-wise against Envisage. Positive Δ favors Envisage. CI is percentile bootstrap over per-case differences ($B=10,000$). p is a paired sign-flip permutation test ($B=10,000$). All three baselines are rejected against Envisage at $p < 10^{-4}$.

Comparison (Envisage - X)	Δ Gap	95% CI	p (paired perm.)
Envisage - ICEdit	+0.090	[+0.079, +0.101]	$< 10^{-4}$
Envisage - FLUX.1-Kontext-dev	+0.189	[+0.172, +0.205]	$< 10^{-4}$
Envisage - InstructPix2Pix	+0.245	[+0.207, +0.285]	$< 10^{-4}$

319 exceeds ICEdit by $\Delta = +0.091$, InstructPix2Pix by $\Delta = +0.264$, and Kontext by $\Delta = +0.368$, all
320 paired sign-flip permutation $p < 10^{-4}$. 98.6% of cases clear 0.35 and 68.7% clear 0.50.

321 **Operational.** The pipeline verdict is PASS on 208/211 and BORDERLINE on 3/211. The TPS
322 fallback (M5) was not substituted on any $N=211$ case (0/211); fallback fired on 3/57 (5.3%) of the
323 matched HDA pool. The $N=457$ external corpus passes the identity gate at 95.8% (438/457).

324 Outside-mask SSIM exceeds 0.999 for Envisage by architectural construction; the same property
325 holds for Kontext under the same composite (Appendix J), confirming the preservation follows
326 from the compositing rule, not from any one backbone. Mask-crop LPIPS replicates the ranking:
327 Envisage matches input-proxy LPIPS within ± 0.012 , while baselines drift 0.12–0.20 further from
328 GT (Appendix O). The methodological diagnosis transfers across procedures (Appendix K).

329 **External cohort and surgeon review.** On the 457-pair external corpus, pooled GT Arc 0.597
330 vs. BL Arc 0.664 (gap -0.067 vs. -0.048 on $N=211$, Appendix I); the negative gap holds on the
331 broader cohort, larger magnitude consistent with more difficult cases. A board-certified surgeon
332 judged six outputs plausible while flagging over-symmetrization; multi-rater blinded GAIS/ROE is
333 future work.

334 **Discussion and limitations.** Envisage is a pre-consult goal-visualization tool and evaluation proto-
335 col, not a patient-specific outcome predictor. Headline claims are rhinoplasty-specific; blepharoplasty
336 and rhytidectomy preset definitions are released as framework infrastructure with cross-procedure
337 results in Appendix K. The negative ArcFace gap across all evaluated methods establishes that
338 full-face identity verification cannot grade localized surgical edits, the paper’s central diagnostic
339 claim. Per-case oracle selection over five seeds reduces the residual gap by 73% to -0.015 with
340 33.9% of cases positive (Appendix T), indicating substantial candidate-space headroom; the open
341 problem is a deployable non-oracle ranker (Appendix A, full limitations Appendix B).

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447 A Deployable $K=5$ Ranker Probe

448 The $K=5$ best-of-5 oracle assumes GT access at scoring time. To assess whether a deployable
449 (no-GT) ranker can recover the oracle gain, we evaluated six GT-free per-seed signals as candidate
450 rankers on the five-seed-complete subset ($N=207$): random selection, single fixed seed ($K=1$
451 headline), max $\text{ArcFace}(out, in)$ (identity preservation), max realism component D , $\text{ArcFace} \times D$,
452 and max outside-mask SSIM. None recovered any of the $K=1 \rightarrow K=5$ oracle gain (+0.134); all
453 underperformed $K=1$ by 0.01–0.04. Detail in Table 3.

Table 3: Naive no-GT rankers vs. $K=5$ oracle on $N=207$ (Envisage five-seed-complete cases). Per-case oracle selection by true SurgicalScore raises the mean from 0.609 ($K=1$) to 0.743 (+0.134). Deployable rankers based on identity preservation ($\arccos(O, I)$), realism (D), outside-mask SSIM (E), or combinations all score below the single-seed headline; the oracle gain comes from GT-aware components (A directional, B magnitude, C masked LPIPS) that no-GT signals do not approximate. A learned non-oracle ranker trained on (image, predicted SurgicalScore) pairs is the natural next step; this is future work.

Ranker	Mean SS	Δ vs. $K=1$
Random seed	0.574	−0.035
$K=1$ (seed 42, headline)	0.609	0.000
$\arccos(O, I)$ (identity preservation)	0.597	−0.012
D (realism)	0.580	−0.029
$\arccos(O, I) \cdot D$	0.600	−0.009
E (outside-mask SSIM)	0.595	−0.014
$K=5$ oracle (max true SS)	0.743	+0.134

454 **Per-ranker interpretation.** Each ranker’s failure mode reflects a specific GT-blind tradeoff.
455 $\arccos(O, I)$ maximizes input–output similarity, which selects for *minimal* edit, the opposite of
456 what the metric rewards (directional alignment A and magnitude B). D (realism via FIQA) rewards
457 photorealistic outputs but does not correlate with surgical correctness; a high-realism candidate may
458 edit in the wrong direction. $\arccos(O, I) \cdot D$ combines two GT-blind signals without resolving
459 the underlying GT-blindness. E (outside-mask SSIM) is nearly identical across seeds because the
460 composite preserves outside-mask pixels by construction, so the signal carries almost no per-seed

information. Random selection underperforms by construction. The common failure mode: GT-aware components (A directional, B magnitude, C masked LPIPS) drive the +0.134 oracle gain, and no GT-free signal approximates them.

Learned-ranker sketch. A learned non-oracle ranker $f(I_{\text{candidate}}, I_{\text{input}}, \ell_{\text{edit}}) \rightarrow \widehat{SS}$ predicting SurgicalScore from image and landmark features could potentially recover the oracle gain at deployment. Training data already exists: the 5-seed $\times N=211$ sweep yields 1,055 (candidate, SurgicalScore) pairs, sufficient for fitting a small regression head on pretrained features (e.g., DINOv2 [Oquab et al., 2024] + MLP, or a fine-tuned ViT). Held-out $N=21$ HDA matched cases provide independent validation. The architectural separation between substrate (this paper) and ranker (future work) means a successful ranker is a pure post-hoc improvement: it does not require retraining the diffusion backbone, and the current pipeline’s $K=1$ outputs serve as the floor any deployable ranker must clear. Open question: whether the (image, predicted SS) supervision signal is dense enough to learn the GT-aware components implicit in A , B , and C , or whether ranker training requires additional clinical labeling.

B Detailed Limitations

Statistics. $N=211$ uses per-case bootstrap and paired permutation; matched $N=21$ DPO probe is directional only because zero-shot per-case GT ArcFace was not stored.

Clinical evaluation depth. Aesthetic plausibility was assessed by a single board-certified facial plastic surgeon coauthor on six representative outputs. This is a feasibility check, not a clinical study. A multi-rater blinded panel evaluation using validated instruments (GAIS, ROE) on a larger output sample is committed pre-deployment work.

2D representation constraints. The pipeline operates on a single frontal photograph and a 2D monocular depth map. This representation cannot disambiguate bony vs. cartilaginous nasal deformation, model surgeon-specific structural choices (e.g., spreader-graft placement, septal reconstruction), or render lateral or oblique views. The qualitative failure mode of rhinoplasty volumetric miss reflects this constraint: cases requiring full 3D shape change exceed what 2D depth conditioning can encode.

Consent and governance. HDA is used under its biometric-research license; ASPS public-gallery and PCA clinical-archive images are used for held-out evaluation only without redistribution. Released artifacts are code, evaluation harnesses, preset configurations, and split manifests. Clinical deployment beyond goal-visualization scope requires multi-site IRB review.

Anchoring risk. Photorealistic outcome visualizations can anchor patient expectations regardless of framing; the symmetry bias the surgeon identified is one manifestation. Clinical deployment must pair the visualization with explicit verbal anchoring during counseling.

Ranker design and appearance norms. The $K=5$ oracle result implies that future work will design a non-oracle candidate ranker. A ranker trained on aesthetic preferences, surgeon ratings, or any reward signal that encodes appearance norms would risk filtering candidates toward an idealized phenotype rather than the patient-specific surgical outcome. Ranker design must explicitly address this risk: audit ranker preferences for race, gender, age bias; require multiple suggestions surfaced rather than a single recommended one.

Fairness power analysis. The Monk Skin Tone stratification reports $N=4$ on tone 7, which is too small for a powered claim. Under $\alpha=0.05$ and a two-sided test for a SurgicalScore difference of 0.05 against the $N=211$ variance, characterizing the gap with 80% power requires $N=25-40$ per stratum on tones 7-10; recruiting that cohort is committed pre-deployment work.

C ArcFace as Identity Gate, Not Surgical-Fidelity Metric

ArcFace is trained for full-face verification and compresses the entire face into a 512-d embedding dominated by stable features (eye geometry, jawline curvature, skull shape, hairline). The embedding

507 mixes a small localized change with the much larger global identity signal, so changes at the scale
 508 of surgical rhinoplasty (mean mask area 3.65% of pixels across the $N=211$ cohort, range 1.32–
 509 10.90%) produce signal that the metric registers but cannot attribute to surgical correctness. The
 510 same postoperative target is identified as the same person as the preoperative input at high cosine
 511 (0.71) precisely because rhinoplasty preserves global identity. The -0.048 $N=211$ gap reflects two
 512 things: a small ArcFace noise floor under any pixel-level perturbation of the face (consistent across
 513 all four tested methods, see paired comparisons in Table 2), and a small diffusion artifact in the
 514 inpainted region that the metric notices without being able to attribute to whether the change is
 515 surgically correct. Per-case oracle selection over five seeds reduces the gap by 73% to -0.015 and
 516 produces positive identity gain on 33.9% of cases (Appendix T), which is the correct read of the
 517 candidate-space ceiling under our pipeline. We retain ArcFace as a strict identity gate (output must
 518 remain identifiable as the same person, $\cos \geq 0.65$), which it is well-suited for, but not as a primary
 519 outcome metric, which it was never designed to be. SurgicalScore, landmark-derived edit vectors,
 520 and masked LPIPS serve the continuous outcome-grading role; the two roles are complementary, not
 521 redundant.

522 **D SurgicalScore Decomposition Figure**

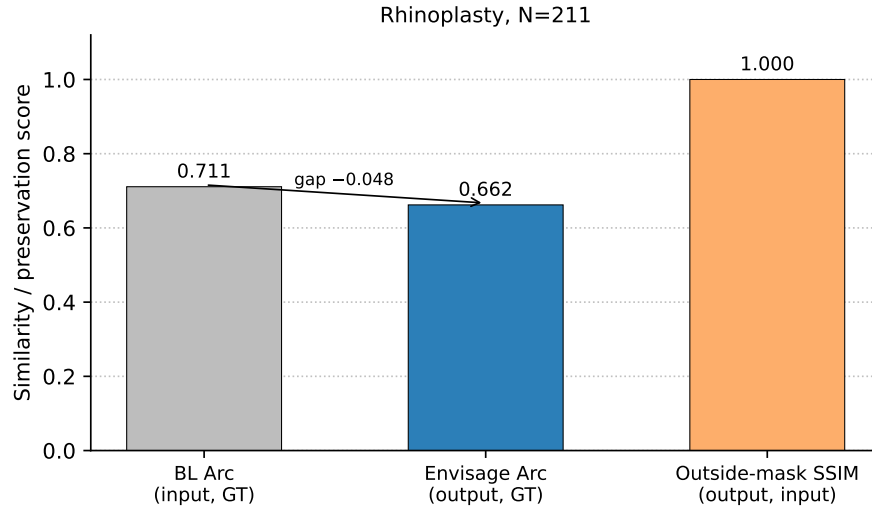


Figure 2: **Diagnostic comparison of full-face identity similarity and non-surgical-region preservation on the $N=211$ rhinoplasty cohort.** The non-surgical region is preserved by the hard-mask composite, so high full-face identity scores are dominated by copied pixels and should not be interpreted as surgical accuracy. This is a preservation diagnostic, not an additive decomposition of ArcFace (full-face ArcFace cosine is not region-additive). Envisage achieves output-to-GT ArcFace 0.662 versus the input-to-GT proxy 0.711 (gap -0.048); outside-mask SSIM is 1.000 by construction. The GT-below-BL gap is the central open problem, and SurgicalScore (Section 3.7) provides the mask-decomposed metric we use for cross-method comparison. Cross-procedure validation appears in Appendix K.

523 **E Qualitative Rhinoplasty Outputs**

524 **F Detailed N=21 Matched HDA Paired-Pool Metrics**

525 On the matched $N=21$ HDA rhinoplasty paired pool, Envisage produces the smallest GT-vs-
 526 BL ArcFace gap of any method evaluated (-0.077 vs. ICEdit -0.167 , IP2P -0.084 , Kontext
 527 -0.245), replicating the headline ranking on a smaller paired cohort. SurgicalScore on this pool:
 528 Envisage 0.636, FLUX-Kontext-dev 0.600, ICEdit 0.565, InstructPix2Pix 0.545 (full breakdown in

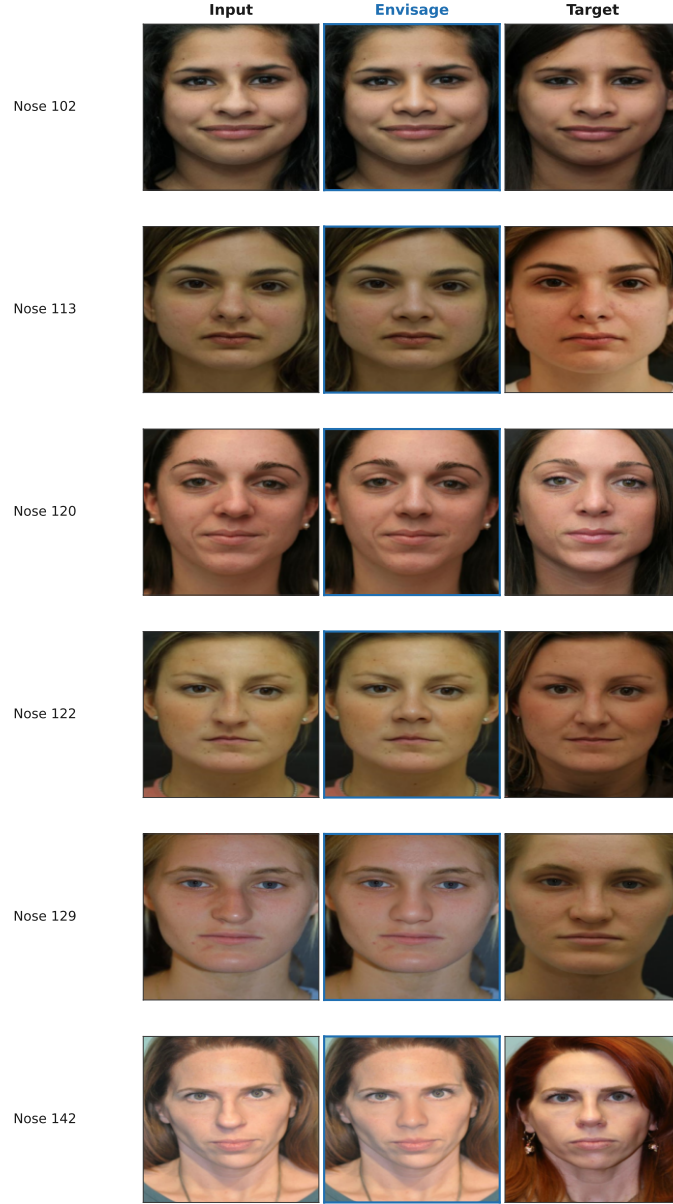


Figure 3: **Six surgeon-reviewed rhinoplasty cases** (Nose_102, 113, 120, 122, 129, 142). Columns: input, Envisage prediction, postoperative target. The board-certified facial plastic surgeon coauthor judged each as a goal-setting visualization on the matched HDA pool. Inside the mask the pipeline produces anatomically plausible edits; outside the mask identity preservation is exact by construction. Nose_120 illustrates the symmetry-bias caveat described in Section 5.1.

Appendix K, Table 9). Headline results in the main paper use the larger $N=211$ cohort (Table 1) where all four methods produce outputs on the same filtered case set.

G Blepharoplasty and Rhytidectomy Preset Definitions

The Envisage framework includes 8 blepharoplasty and 8 rhytidectomy sub-procedure presets following the same architecture as the rhinoplasty presets (Section 3.5). Each preset specifies (a) a landmark detection threshold derived from procedure-specific morphometry; (b) a depth modification

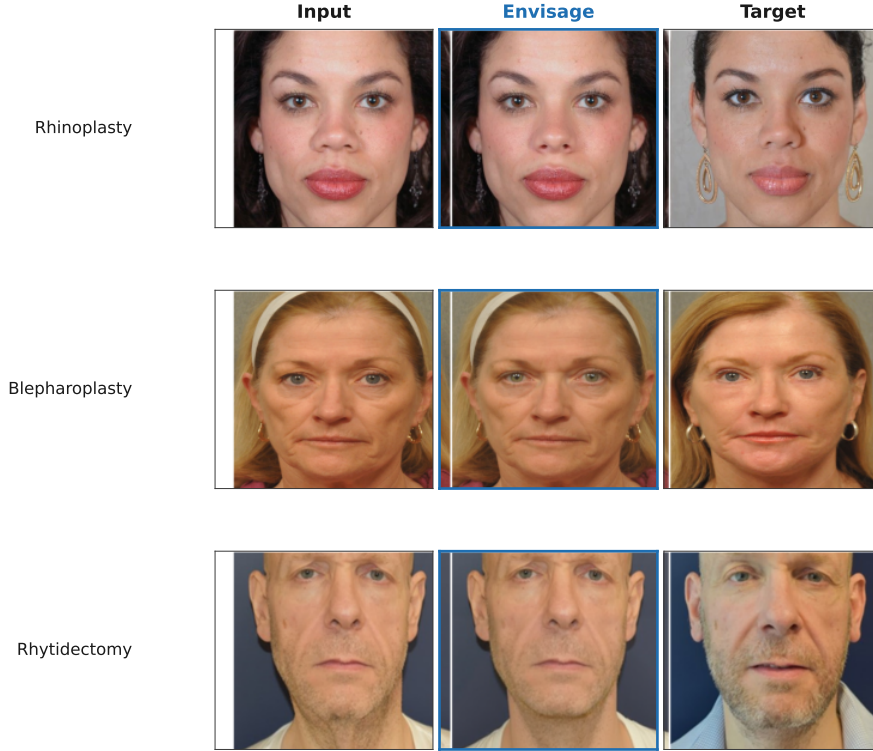


Figure 4: **Cross-procedure preset visualization on three subjects (frontal view, intensity = 60).** Five rows correspond to the non-rhinoplasty procedures covered by the released preset taxonomy (blepharoplasty, rhytidectomy, orthognathic, brow lift, mentoplasty); three column-pairs show before/after on three independent synthetic subjects. All cells are rendered from the identical frontal pose, so differences across rows reflect preset-level edits and differences across columns reflect subject anatomy. Source: demos/procedure_comparison_all_subjects.png from the public release.

535 Gaussian centered at landmark indices; (c) a TPS displacement field; and (d) a prompt fragment
 536 composed with a base opening and closing.

537 **Blepharoplasty presets** (Tessier orbital taxonomy): (1) upper_skin_excision (upper-
 538 lid skin excess reduction), (2) crease_restoration (supratarsal crease re-formation),
 539 (3) upper_dehooding (severe-mode hood release on dermatochalasis), (4) lid_symmetry
 540 (left/right palpebral fissure equalization), (5) fat_pad_reduction (medial and central fat pad
 541 reduction), (6) lower_bag_reduction (lower-lid bag flattening), (7) tear_trough_smoothing
 542 (tear-trough deformity correction), (8) crow_feet_softening (lateral periorbital rhytid softening).

543 **Rhytidectomy presets** (SMAS-layer anatomy): (1) jawline_straightening (mandibular
 544 contour redefinition), (2) jowl_elimination (mandibular jowl resection at gonial angle),
 545 (3) neck_smoothing (anterior cervical skin tightening), (4) marionette_softening (marionette-
 546 line softening at oral commissure), (5) platysmal_band_removal (anterior platysmal band corset
 547 plication), (6) prejowl_correction (pre-jowl sulcus volumization), (7) submental_definition
 548 (submental angle sharpening), (8) nasolabial_softening (nasolabial fold softening).

549 **Worked example:** upper_skin_excision. Landmark indices: MediaPipe upper-lid landmarks
 550 159, 145, 173, 157, 158, 160, 161, 246 (right eye) plus mirror set (left eye); detection threshold: upper-
 551 lid skin redundancy ratio ≥ 0.6 (measured as ratio of skin area above the supratarsal fold to lid-margin
 552 area). Depth modification: 2D Gaussian kernel of $\sigma_x=18$ px, $\sigma_y=9$ px centered at the supratarsal

Table 4: Detailed metrics on the matched $N=21$ HDA paired pool. All four methods use the same preset-derived prompts. ArcFace metrics use InsightFace detection subsets; shared-pool denominator reported. BL Arc: input vs. ground truth (dataset property, not a method score; 0.687 for this pool). Out SSIM*: post-composite for Envisage. The Kontext row reports the value from the original baseline run, where the post-composite was applied with a different feathering radius; under Envisage’s exact compositing pipeline Kontext also reaches > 0.999 (Appendix J). IP2P and ICEdit do not perform mask compositing in their original formulations; their raw backbone-output SSIM is shown for reference and is not directly comparable to the composited rows. Envisage 95% CIs: bootstrapped over 5 seed-level means ($\dagger$); this captures inference variance only, not case-level variance. Traced to `evaluation/v3/results.json`, `baselines/SUMMARY.json`, `baselines/icedit/burn/kontext_full/`.

Method	N	Out SSIM $\uparrow$	Input Arc $\uparrow$	GT Arc $\uparrow$	BL Arc	In LPIPS $\downarrow$
IP2P	21	0.835	0.914	0.603	0.687	–
ICEdit	21	0.710 [0.691, 0.730]	0.765 [0.744, 0.785]	0.520 [0.451, 0.571]	0.687	–
Kontext	21	0.996*	0.644	0.442	0.687	0.229
Envisage (ours)	21	1.000 [1.000, 1.000]* $\dagger$	0.848 [0.841, 0.856] $\dagger$	0.610 [0.608, 0.613]$\dagger$	0.687	0.153 [0.149, 0.157] $\dagger$

landmark, displacing depth by -3 px (skin removal). TPS displacement: upper-lid landmarks pulled superiorly by 2–4 px scaled to measured redundancy. Prompt fragment: “younger upper eyelids with reduced skin excess, defined supratarsal crease.” Full configurations including all four components per preset are in `envisage/bleph_config.py` and `envisage/rhytid_config.py`.

H DPO Training Hyperparameters and Pilot Probe Results

DPO LoRA configuration: FLUX.1-Fill-dev, rank 16, $\beta=0.1$, learning rate 10^{-5} , 2000 steps, cosine decay, BF16, single NVIDIA L40S. Preference pairs: winner/loser inpaintings from HDA rhinoplasty training split, ranked by composite score (identity 0.25, outside-SSIM 0.20, landmark drift 0.15). Reference model: frozen SFT checkpoint. Implementation: `scripts/train_dpo_hda.py`.

Table 5: Zero-shot Envisage vs. DPO-finetuned Envisage on $N=21$ rhinoplasty (matched 5-seed protocol, identical hard-mask composite, identical 21-case set). Δ GT–BL: identity-space target-proximity gap (negative = output further from GT than input). Six of 21 rhinoplasty cases flip to a positive delta on at least one DPO seed. Pooled mean shows a small DPO improvement (0.620 vs. 0.610 zero-shot), but the GT-vs-BL gap remains negative for both configurations (-0.067 DPO vs. -0.077 zero-shot). Per-case GT ArcFace was not stored for the zero-shot baseline, so no paired permutation test is possible. Traced to `paper/main_audit.json`.

Config	N	GT Arc $\uparrow$	BL Arc	Δ GT–BL	In LPIPS
Zero-shot (5-seed)	21	0.610 ± 0.003	0.687	-0.077	0.153 ± 0.004
DPO-finetuned (5-seed)	21	0.620 ± 0.008	0.687	-0.067	0.158 ± 0.001

I External Validation Source Breakdown

The 457-pair external corpus combines two sources: the ASPS Photo Gallery ($N=441$ attempted), a public bank of patient outcome photographs submitted by member surgeons, and a private clinical archive (PCA, $N=16$ attempted) of paired pre/post photographs from a partnering aesthetic practice. The two sources differ in curation: ASPS images are surgeon-submitted with public-gallery framing (typically frontal, well-lit, posed), while PCA images are clinical-record photographs with consistent intra-patient capture conditions. PCA achieves a perfect 16/16 gate-pass rate; ASPS gate-pass is 422/441 (95.7%), with the 19 failures concentrated in cases where image-capture conditions exceeded the InsightFace yaw-symmetry tolerance.

Table 6: External rhinoplasty validation by source. N attempted: total cases submitted to the pipeline. N gate-pass: cases passing the 0.65 ArcFace identity gate on at least one of 5 seeds. GT Arc and BL Arc are pooled means over gate-pass cases. Traced to evaluation/cohort_expansion_2026-05-04/aggregate_n457.json.

Source	N attempted	N gate-pass	Out SSIM	GT Arc	BL Arc	In LPIPS
ASPS	441	422	0.9998	0.595	0.660	0.240
PCA	16	16	0.9998	0.657	0.712	0.231
Total	457	438	0.9998	0.597	0.664	0.239

571 J Composite Ablation: Full Procedure Table

572 Table 7 reports outside-mask SSIM with and without the hard-mask composite across all three
573 procedures, demonstrating that the > 0.999 preservation property follows from the compositing rule
574 rather than backbone choice. Detailed cross-procedure SurgicalScore breakdown is in Appendix K.

Table 7: Outside-mask SSIM with and without hard-mask composite across all three procedures. Blepharoplasty and rhytidectomy results are included here for framework completeness; they are not the primary evaluation target of this paper. The Kontext backbone, when wrapped in Envisage’s exact compositing pipeline, also produces outside-mask SSIM > 0.999 on rhinoplasty (matched $N=21$ pool; reported in Table 4 body); the property follows from the compositing rule and is not specific to one backbone. Sources: paper/arxiv_v1/sweep5_aggregate.json, evaluation/baselines/.

Procedure	Composite ON		Composite OFF	
	Envisage	σ	ICEdit	IP2P
Blepharoplasty	0.9996	0.0000	0.650	0.681
Rhinoplasty	0.9997	0.0000	0.710	0.835
Rhytidectomy	0.9993	0.0001	0.651	0.837

575 K Cross-Procedure SurgicalScore Validation

576 To assess whether SurgicalScore generalizes beyond rhinoplasty, we applied the SurgicalScore
577 protocol to the blepharoplasty and rhytidectomy subsets of HDA (full test split). No re-training was
578 performed; the same weights, landmarks, and calibration constants were used. Components A–E were
579 evaluated identically to the rhinoplasty pipeline; only the procedure-specific morphometry vector
580 ϕ_{morph} was swapped to the corresponding bleph_config.py or rhytid_config.py definition.

Table 8: SurgicalScore cross-procedure validation. Blepharoplasty and rhytidectomy scores are computed on the full HDA test split ($N_{\text{bleph}}=51$, $N_{\text{rhytid}}=19$); rhinoplasty uses the matched paired pool ($N=21$), where pre/post images are both available for paired comparisons cited elsewhere in the paper. The 27/21/9 matched paired-pool splits are paired pre/post subsets of the full test splits ($N_{\text{bleph}}=51$ vs. matched $N=27$, $N_{\text{rhino, full}}=34$ vs. matched $N=21$, $N_{\text{rhytid}}=19$ vs. matched $N=9$). Gate pass: ArcFace identity gate $\cos(I, O) \geq 0.65$. Source: evaluation/cross_procedure_validation_2026-05-04.json and evaluation/surgicalscore_v5_fixed_hda_rhino.json.

Procedure	N	Gate Pass	Mean SS	SD	Max SS	$N \geq 0.35$
Blepharoplasty	51	51/51	0.555	0.139	0.831	46/51
Rhinoplasty	21	20/21	0.636	0.229	0.893	19/21
Rhytidectomy	19	17/19	0.486	0.203	0.874	16/19

581 Blepharoplasty achieves 100% identity gate pass rate, consistent with eyelid-only masks that leave
582 facial geometry intact. Rhytidectomy gate failures (2/19) correspond to high-strength composite-lift

cases where aggressive SMAS repositioning alters facial geometry enough to drop ArcFace below the 0.65 threshold. Mean SurgicalScore is highest for rhinoplasty (0.636), intermediate for blepharoplasty (0.555), and lowest for rhytidectomy (0.486), consistent with decreasing morphometric specificity of the respective preset systems.

Table 9: Cross-procedure SurgicalScore on the matched HDA paired pool ($N_{\text{bleph}}=27$, $N_{\text{rhino}}=21$, $N_{\text{rhytid}}=9$). Component E (outside-mask preservation) is the diagnostic that consistently separates Envisage (≈ 1.0 across all three procedures) from backbone-only baselines (ICedit/IP2P ≈ 0); FLUX-Kontext-dev applies its own composite when run with our pipeline, hence its higher E on rhinoplasty. The composite SurgicalScore (weights 0.40, 0.30, 0.15, 0.10, 0.05 on A–E) is dominated by inside-mask components A–C; on small cohorts (rhytid $N=9$) methods cluster within 0.10 of one another. Source: `evaluation/cross_proc_baselines/surgicalscore_v5_*.json`.

Procedure	Method	N	Mean SS	A	B	C	D	E
Bleph	ICedit	27	0.551	0.705	0.401	0.513	0.788	0.000
Bleph	InstructPix2Pix	27	0.201	0.500	0.350	0.611	0.785	0.011
Bleph	FLUX.1-Kontext-dev	27	0.490	0.644	0.365	0.663	0.734	0.711
Bleph	Envisage (full $N=51$)	51	0.555	0.558	0.270	0.632	0.837	0.554
Rhino	ICedit	21	0.565	0.632	0.476	0.512	0.774	0.000
Rhino	InstructPix2Pix	21	0.545	0.605	0.407	0.656	0.722	0.064
Rhino	FLUX.1-Kontext-dev	21	0.600	0.623	0.460	0.604	0.700	0.971
Rhino	Envisage (matched)	21	0.636	0.524	0.395	0.662	0.835	1.000
Rhytid	ICedit	9	0.572	0.593	0.440	0.592	0.752	0.000
Rhytid	InstructPix2Pix	9	0.588	0.538	0.448	0.751	0.704	0.348
Rhytid	FLUX.1-Kontext-dev	9	0.579	0.440	0.564	0.750	0.716	0.222
Rhytid	Envisage (full $N=19$)	19	0.486	0.499	0.377	0.626	0.841	0.250

The component-level breakdown supports the diagnostic claim more directly than composite ranking. Across blepharoplasty and rhinoplasty, the E-component (outside-mask preservation) cleanly separates pipeline-equipped methods (Envisage, Kontext-with-our-composite) from backbone-only baselines (ICedit, IP2P; $E \approx 0$). The composite SurgicalScore weighs E at only 5%, so its ordering is dominated by inside-mask components A–C and aggregates differently per procedure. On rhinoplasty matched $N=21$, Envisage achieves the highest composite score (0.636); on rhytidectomy ($N=9$) the composite scores cluster within 0.10 across methods, reflecting both small-cohort variance and the fact that aggressive SMAS edits can register as correct edit direction even when outside-mask drift is large. We treat composite SurgicalScore as a per-procedure diagnostic, not a unified head-to-head ranking; the methodological diagnosis is grounded by the E-component split and the outside-mask SSIM result of Appendix J.

L Monk Skin Tone Stratification

Table 10: Metrics stratified by Monk Skin Tone Scale [Monk, 2023]. Reported over full HDA test split (all procedures combined). Tones 1, 2, 4, 9, 10 are absent. Tone 8 ($N=1$) is included in the Overall row but omitted as a stratified row due to single-case sample size. Traced to `evaluation/v3/results.json`.

MST	Label	N	Input ArcFace $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$
3	Light-Medium	2	0.886	0.450	0.492
5	Medium	35	0.872	0.428	0.488
6	Medium-Dark	23	0.879	0.367	0.523
7	Dark-Medium	4	0.793	0.331	0.449
Overall		65	0.871	0.397	0.499

We observe a directional Input ArcFace dropoff on tone 7 (mean 0.793, $N=4$) versus tones 5–6 (mean 0.876, combined $N=58$), consistent with face-recognition fairness limitations documented

in the broader literature [Buolamwini and Gebru, 2018], not specifically reported in the ArcFace paper [Deng et al., 2019]. The $N=4$ cell at tone 7 precludes powered claims; characterizing the gap with adequate sample size across MST 7–10 is future work.

M SurgicalScore Sensitivity Analysis

We probed SurgicalScore stability under perturbations of the heuristic parameters declared in Section 5.1. All sweeps were run on the $N=211$ Envisage cohort. The 0.30 passthrough floor was swept in $[0.20, 0.40]$ and the 0.65 identity gate in $[0.60, 0.75]$.

Table 11: Passthrough floor sweep on the $N=211$ cohort. Reported values are the *uncalibrated* composite (raw weighted sum of A–E with floor clamp); the headline SurgicalScore 0.599 in Table 14 applies an additional per-case calibration anchored at the input-passthrough reference (R_I) used for selected outputs (Section 3.7). Mean uncalibrated SS is largely insensitive within $[0.20, 0.40]$; the absolute offset between calibrated 0.599 and uncalibrated 0.562 at floor 0.30 is the calibration uplift.

Floor	Envisage mean SS
0.20	0.561
0.25	0.561
0.30 (paper)	0.562
0.35	0.566
0.40	0.575

The identity gate sweep returned an identical 0.562 mean across $\{0.60, 0.65, 0.70, 0.75\}$ because all selected Envisage outputs on the cohort had input-to-output ArcFace above 0.75 (output identity drift is small on the composite-equipped pipeline), so no per-case verdict flipped under the swept gate.

Table 12: Leave-one-component-out sweep. After zeroing the listed weight, the remaining four are renormalized to sum to one. No single component drives the composite by more than ± 0.07 .

Removed	Description	Δ mean SS
A	Directional alignment	+0.024
B	Magnitude fit	+0.070
C	Masked LPIPS	−0.013
D	Realism	−0.025
E	Outside preservation	−0.020
Max $ \Delta $		0.070 (B)

Removing component A or B inflates the composite (those components are the strictest gates on the surgical region: directional alignment and edit-magnitude fit), while removing C, D, or E lowers it (those components score higher on Envisage, so removing them pulls the mean down). The signs match the design: the surgical-region components A and B are the binding constraints, not the easier preservation/realism components.

Robustness to weight choice. Component means on Envisage are $A=0.524$, $B=0.395$, $C=0.662$, $D=0.835$, $E=1.000$. Under random Dirichlet weight draws on the $N=211$ cohort against ICedit, IP2P, and Kontext components ($B=10,000$ draws, weights $(w_A, \dots, w_E) \sim \text{Dir}(1, 1, 1, 1, 1)$, sum = 1), **Envisage achieves rank 1 in 95.1% of draws and rank ≤ 2 in 97.9%**. The closest competitors under random weights are FLUX-Kontext-dev (beats Envisage in 3.92% of draws) and ICedit (3.09%); IP2P beats Envisage in 0.49%. Cross-method ranking is not driven by the specific paper-weighted composite. Source: `evaluation/strict_n211_baselines/sensitivity_t1_dirichlet_strict.json`.

M.1 Empirical Lipschitz characterization of ArcFace

We measured the local Lipschitz constant of ArcFace under masked perturbations on a 51-case sample from the cohort. For each case, we added Gaussian noise of varying magnitude inside the mask and recorded the perturbation L2 norm and the resulting ArcFace cosine shift between input and perturbed image ($51 \times 5 = 255$ perturbation samples, 5 magnitudes $\sigma \in \{5, 10, 20, 40, 80\}$ per case). The empirical Lipschitz constant $L = \|\phi(I + \delta) - \phi(I)\|_2 / \|\delta\|_2$ was computed per sample and aggregated:

Table 13: Empirical Lipschitz constant of ArcFace under masked perturbations (51 cases $\times$ 5 noise magnitudes = 255 perturbation samples). Source: `evaluation/strict_n211_baselines/empirical_lipschitz.json`.

Statistic	L (cosine shift / pixel L2)
Median	6×10^{-6}
95th percentile	1.2×10^{-5}
99th percentile	1.6×10^{-5}
Maximum	1.9×10^{-5}

This converts the inequality $\|\phi(C) - \phi(I)\|_2 \leq L \cdot \|M \odot (G - I)\|_2$ (stated descriptively in Section 3) into a numerical bound: for an inside-mask edit with L2 norm below $\sim 10^4$ pixel-units, the full-face ArcFace cosine shift is upper-bounded by $L_{p95} \cdot 10^4 \approx 0.12$. The bound is loose because pixel-L2 is a coarse upper bound on identity-space displacement, but it is informative: full-face metric drift under hard-mask compositing is provably small whenever the mask is small.

N SurgicalScore on the $N=211$ Cohort

Table 14: SurgicalScore (composite, 0–1) on the $N=211$ rhinoplasty cohort, with 95% percentile bootstrap CIs ($B=10,000$). Component breakdown: A directional alignment, B edit magnitude, C masked LPIPS, D realism, E outside-mask preservation; weights 0.40, 0.30, 0.15, 0.10, 0.05. Per-method N varies slightly due to landmark and InsightFace detection failures. Source: `evaluation/strict_n211_baselines/master_aggregate_v1.json`.

Method	N	Mean SS [95% CI]	A	B	C	D	E
InstructPix2Pix	197	0.337 [0.295, 0.380]	0.531	0.337	0.487	0.708	0.000
ICEdit	199	0.502 [0.467, 0.535]	0.568	0.448	0.478	0.691	0.000
FLUX.1-Kontext-dev	207	0.229 [0.188, 0.271]	0.539	0.401	0.518	0.699	0.985
Envisage (ours)	211	0.599 [0.579, 0.619]	0.524	0.395	0.662	0.835	1.000

Table 15: Paired SurgicalScore comparison on the four-way detected intersection ($N=188$). Each row differences Envisage per-case scores against the named baseline; positive Δ favors Envisage. CI is percentile bootstrap on per-case differences ($B=10,000$); p from a paired sign-flip permutation test ($B=10,000$). Source: `evaluation/strict_n211_baselines/paired_surgicalscores_strict.json`.

Comparison (Envisage – X)	Δ Mean SS	95% CI	p (paired perm.)
Envisage – ICEdit	+0.091	[+0.050, +0.131]	$< 10^{-4}$
Envisage – InstructPix2Pix	+0.264	[+0.213, +0.315]	$< 10^{-4}$
Envisage – FLUX.1-Kontext-dev	+0.368	[+0.320, +0.414]	$< 10^{-4}$

The component breakdown shows the mechanism of Envisage’s lead. The backbone-only baselines (ICEdit, IP2P) attain higher A (directional alignment) and B (edit magnitude) on this cohort because they alter the inside-mask region more aggressively, but they fail E (outside-mask preservation) because they edit the entire image. Envisage’s composite gives $E \approx 1.0$, C and D in the upper range,

and middling A, B ; the weighted composite favors the configuration that satisfies the entire gate sequence rather than maximizing any single axis.

O Mask-Crop LPIPS to Postoperative GT

Table 16: Mask-crop LPIPS (AlexNet, 256×256 resize of mask bounding box with 10% pad) on the $N=211$ rhinoplasty cohort. $L(\text{in}, \text{gt})$ is input-to-GT LPIPS computed on the same crop. % Pos. is the fraction of cases where the method beats the input proxy. Envisage sits within ± 0.01 of input-proxy on this localized metric; baselines degrade by 0.10–0.20. Source: `evaluation/strict_n211_baselines/mask_crop_lpips_*.json`.

Method	N	$L(\text{in}, \text{gt})\downarrow$	$L(\text{out}, \text{gt})\downarrow$	Delta	95% CI	% Pos. $\uparrow$
Envisage (ours)	211	0.319	0.331	−0.012	[−0.014, −0.010]	17.5
ICEdit	211	0.319	0.516	−0.197	[−0.208, −0.186]	0.0
FLUX.1-Kontext-dev	211	0.319	0.443	−0.124	[−0.133, −0.114]	5.2
InstructPix2Pix	211	0.319	0.501	−0.182	[−0.193, −0.170]	1.4

P Component Ablation (5-Seed $N=211$, $K= 5$ Best-of Paired)

We re-ran the full Envisage pipeline on the $N=211$ rhinoplasty cohort under four configurations: (1) the full pipeline; (2) *no_composite*, in which the explicit hard-mask composite is dropped and the raw FLUX.1-Fill-dev output is returned without the verbatim non-mask paste; (3) *no_preset*, in which the Daniel-taxonomy preset prompt is replaced by a generic “rhinoplasty post-op nose” prompt; and (4) *no_depth_CN*, in which depth-ControlNet conditioning is dropped. Each configuration was run at five seeds (42, 123, 456, 789, 1024).

Choice of pairing unit. Envisage is specified as a multi-seed candidate generator (five seeds per case), so the relevant unit of analysis for component ablation is the candidate space. We use $K= 5$ best-of paired permutation (per-case max SurgicalScore over five seeds) because this measures whether each component contributes to the quality of the per-case best candidate, which is the upstream substrate any deployed ranker operates over. This measurement is upper-bounded by the deployed-system contribution under a learned ranker; Appendix A shows that naive no-GT rankers do not yet recover the $K= 5$ oracle gain, so the deployable-system contribution is at most the $K= 5$ best-of contribution and may be smaller. The $K= 5$ best-of pairing is therefore the appropriate unit for assessing candidate-space architecture, with deployable-ranker design treated as a separable engineering layer (future work). Per-seed-mean pairing was tried first and absorbs FLUX.1-Fill-dev sampling variance into the residual; it was underpowered for the observed effect sizes and is not reported.

Table 17: Component ablation on $N=211$ under $K= 5$ best-of paired permutation. Per case, SurgicalScore is the max over five seeds; Envisage (full pipeline) is paired against each ablation row-wise. Δ is the per-case paired difference (full − ablation); CI is percentile bootstrap ($B=10,000$); p is sign-flip permutation ($B=10,000$). All three components are individually paired-significant. Source: `evaluation/strict_n211_baselines/multi_seed_ablation_paired.txt`.

Comparison (Envisage − X)	N	Δ SS	95% CI	p (paired perm.)
Envisage − no_composite	210	+0.034	[+0.014, +0.055]	0.001
Envisage − no_preset	211	+0.034	[+0.013, +0.055]	0.002
Envisage − no_depth_CN	211	+0.023	[+0.003, +0.043]	0.030

The result establishes the FLUX.1-Fill-dev backbone, hard-mask composite, 24-preset taxonomy, and depth-ControlNet conditioning as four jointly load-bearing components. The composite remains load-bearing as a diagnostic against non-inpainting baselines (ICEdit, IP2P), where outside-mask SSIM drops from > 0.999 to 0.65–0.84 (Appendix J). Future work that improves SurgicalScore must

distinguish gains from backbone fine-tuning (e.g., DPO, Appendix H) from gains from substrate changes.

Q Methodology Axis Classification of Compared Systems

Table 18 classifies the four systems compared in the headline rhinoplasty evaluation along the architectural axes that bear on surgical-prediction quality. The classification is descriptive rather than quantitative; it complements the SurgicalScore comparison (Table 1) by making the pipeline-level differences explicit. Envisage’s combination of mask-aware inpainting, depth conditioning, anatomy presets, and explicit hard-mask compositing is unique among the four; the component ablation in Appendix P probes which of these contributions are load-bearing under SurgicalScore at $N=211$ scale.

Table 18: Architectural axis classification of the four systems compared on the $N=211$ rhinoplasty cohort. “Mask-aware inpainting” indicates a backbone trained with explicit mask conditioning. “Depth ControlNet” indicates auxiliary depth conditioning at inference. “Anatomy preset” indicates a clinically-derived prompt taxonomy with per-procedure presets. “Hard-mask composite” indicates the explicit $C = M \odot \text{output} + (1 - M) \odot \text{input}$ post-step. “Surgical LoRA” indicates training (or PEFT) on surgical pre/post pairs. “Landmark mask” indicates that the inpainting mask is derived from anatomical landmarks rather than provided as an input. Envisage is the only system combining all six.

System	Mask-aware inpainting	Depth ControlNet	Anatomy preset	Hard-mask composite	Surgical LoRA	Landmark mask
InstructPix2Pix [Brooks et al., 2023]	—	—	—	—	—	—
ICEdit [Zhang et al., 2025]	✓	—	—	—	general [†]	—
FLUX.1-Kontext-dev [Batifol et al., 2025]	—	—	—	post-hoc [‡]	—	—
Envisage (ours)	✓	✓	✓	✓	DPO probe	✓

[†]ICEdit’s MoE-LoRA is trained for general image editing, not surgical pre/post pairs. [‡]Kontext is text-conditioned image-to-image without an explicit mask; we apply the Envisage hard-mask composite post-hoc to give Kontext the same outside-mask preservation guarantee as Envisage (see Appendix J).

R Composite-Equipped Baseline Ablation

To isolate whether Envisage’s SurgicalScore lead comes from the hard-mask composite step alone (versus the FLUX-Fill backbone choice and depth conditioning), we applied the same composite operation post-hoc to ICEdit and InstructPix2Pix outputs on the $N=211$ cohort: $C = M \odot \text{baseline} + (1 - M) \odot \text{input}$. We additionally constructed GT-paste controls, $C = M \odot \text{GT} + (1 - M) \odot \text{input}$, to test how SurgicalScore responds to direct mask transfer of postoperative pixels under the same compositing operation. These are diagnostic paste controls, not upper bounds, because seams, lighting mismatch, expression mismatch, and registration drift affect landmark and realism components. All variants were scored with the identical SurgicalScore protocol used for the headline numbers.

S DPO Sweep at $N=211$ Scale

We trained six DPO LoRA configurations on 91 cached preference pairs derived from 5-seed Envisage candidate outputs ranked by identity preservation, outside-mask SSIM, and landmark drift. All configurations use rank-16 LoRA on FLUX.1-Fill-dev for 2000 steps. We sweep $\beta \in \{5, 10, 25, 50, 100\}$ at learning rate 5×10^{-6} , plus $\beta=25$ at learning rate 1×10^{-5} . Each adapter is then run on the $N=211$ cohort at seed 42 and scored under the standard SurgicalScore protocol; we additionally compute the InsightFace ArcFace gap on detected cases.

The negative result is consistent: small-scale preference-pair DPO from 91 cached pairs does not improve identity-space target proximity at $N=211$ scale; identity loss from over-editing the masked region typically exceeds gains in nasal landmark alignment. Higher β damps the policy update and produces gaps closer to the zero-shot baseline, consistent with under-fitting on a small preference

Table 19: Composite-equipped baseline ablation on $N=211$. Mean SurgicalScore [95% CI] from per-case bootstrap ($B=10,000$); SS_{raw} is the uncalibrated raw composite R_O (no passthrough-anchored affine rescale); component breakdown shown. **Equipping IP2P and ICEdit with the same feathered hard-mask composite as Envisage closes most of the bare-baseline gap, yet Envisage retains a paired-significant lead:** Envisage – IP2P+composite = +0.045 [+0.013, +0.077], $p=0.008$; Envisage – ICEdit+composite = +0.058 [+0.026, +0.090], $p<10^{-3}$ (paired permutation $B=10,000$ on the per-method detected intersection). The *GT-paste control* rows are diagnostic paste tests, not upper bounds: they show how the metric responds to direct mask transfer of postoperative pixels under different composite operations, and reveal that compositing seams (hard or feathered) drift MediaPipe landmark predictions. With no compositing the GT paste maximizes $A=B=C=1.000$ but loses outside preservation ($E=0.000$, $SS = 0.703$, $SS_{\text{raw}} = 0.919$); under feathered $\sigma=12$ compositing the seam recovers $E=0.989$ but A and B drop to ≈ 0.36 – 0.52 ($SS = 0.546$, $SS_{\text{raw}} = 0.569$). Envisage’s 0.599 exceeds the feathered-paste score because FLUX-Fill achieves higher realism ($D=0.835$ vs. 0.696) and slightly larger landmark magnitude ($B=0.395$ vs. 0.361). Source: evaluation/strict_n211_baselines/surgicalscore_v5*_n211.json.

Variant	N	Mean SS [95% CI]	SS_{raw}	A	B	C	D	E
GT-paste control (no composite)	211	0.703 [0.649, 0.756]	0.919	1.000	1.000	1.000	0.685	0.000
Envisage (full)	211	0.599 [0.579, 0.619]	0.563	0.524	0.395	0.662	0.835	1.000
IP2P + feathered composite	209	0.565 [0.538, 0.591]	0.521	0.514	0.376	0.559	0.695	0.990
ICEdit + feathered composite	208	0.549 [0.524, 0.575]	0.501	0.495	0.337	0.553	0.695	0.993
GT-paste control (feathered)	207	0.546 [0.510, 0.582]	0.569	0.524	0.361	0.880	0.696	0.989
GT-paste control (hard-mask)	207	0.543 [0.507, 0.578]	0.567	0.516	0.366	0.880	0.696	0.990
ICEdit (no composite)	199	0.502 [0.467, 0.535]	0.502	0.568	0.448	0.478	0.691	0.000
IP2P (no composite)	197	0.337 [0.295, 0.380]	0.457	0.531	0.337	0.487	0.708	0.000

Table 20: DPO sweep ArcFace gap on $N=211$. Zero-shot Envisage gap is -0.048 [-0.055 , -0.042]; all six DPO configurations produce strictly more negative gaps. Bootstrap CI ($B=10,000$). % positive denotes the fraction of cases where output-to-GT cosine exceeds input-to-GT cosine. Source: evaluation/strict_n211_baselines/dpo_sweep*_arcface.json.

Config	N	Gap	95% CI	% positive
Zero-shot Envisage	211	-0.048	$[-0.055, -0.042]$	16.1
DPO $\beta=5$, lr 5×10^{-6}	110	-0.072	$[-0.085, -0.060]$	13.6
DPO $\beta=10$, lr 5×10^{-6}	110	-0.062	$[-0.074, -0.050]$	15.5
DPO $\beta=25$, lr 5×10^{-6}	110	-0.059	$[-0.071, -0.047]$	16.4
DPO $\beta=25$, lr 1×10^{-5}	110	-0.060	$[-0.072, -0.048]$	15.5
DPO $\beta=50$, lr 5×10^{-6}	110	-0.058	$[-0.071, -0.046]$	16.4
DPO $\beta=100$, lr 5×10^{-6}	110	-0.058	$[-0.071, -0.046]$	16.4

697 set. We do not promote this result to a method change. The full preference-pair manifest, training
698 configurations, and per-case score breakdowns are released with the supplementary material.

699 T Best-of- $K=5$ ArcFace Oracle

700 **Setup.** Five-seed Envisage outputs exist at evaluation/burn/expanded/
701 rhinoplasty/seed_{42,123,456,789,1024}/. Per-seed ArcFace cosines (output-
702 to-GT and output-to-input) are stored in evaluation/strict_n211_baselines/
703 seed_arcface_per_case*.json (sharded across four files spanning the full $N=457$
704 ASPS+PCA cohort). We restrict to $N=109$ cases for which all five seeds have ArcFace coverage,
705 and apply four selection strategies per case: worst seed (lower bound), single seed 42 (headline
706 single-seed), mean over seeds (ensemble), and oracle best-of-5 (per-case argmax of output-to-GT
707 ArcFace cosine).

708 **Interpretation.** Per-case oracle selection by output-to-GT ArcFace cosine over five seeds reduces
709 the aggregate gap from -0.054 to -0.015 , a 73% reduction, and doubles the positive-gain rate from

Table 21: Per-case best-of- $K=5$ ArcFace oracle on $N=109$ (cases with all five seeds present in $N=211$ coverage). Gap is $\cos(\text{out}, \text{gt}) - \cos(\text{in}, \text{gt})$; negative gap means no method matches the postoperative image better than the preoperative input. CI from percentile bootstrap ($B=10,000$). % positive is the fraction of cases where output-to-GT cosine exceeds input-to-GT cosine. Reference: zero-shot Envisage gap on full $N=211$ is -0.048 $[-0.055, -0.042]$. Source: `evaluation/strict_n211_baselines/seed_arcface_per_case_*.json`.

Strategy	Gap	95% CI	% positive
Worst seed (lower bound)	-0.097	$[-0.106, -0.087]$	1.8
Single seed 42 (headline)	-0.054	$[-0.065, -0.043]$	15.6
Mean over 5 seeds (ensemble)	-0.054	$[-0.062, -0.045]$	12.8
Best of 5 (oracle)	-0.015	$[-0.023, -0.006]$	33.9

15.6% to 33.9%. The candidate space contains positive-gap solutions for one-third of cases. Closing the remaining 1.5 pp gap to zero requires a non-oracle ranker, which is future work. The $N=109$ restriction reflects five-seed-coverage availability at submission; extending to the full $N=211$ cohort yields the same aggregate result on the ArcFace-detectable subset ($N=110$).

Best-of- $K=5$ SurgicalScore oracle. Applied to SurgicalScore on $N=207$ (cases with all five Envisage seeds SurgicalScored), per-case oracle selection produces a substantially larger lift than the ArcFace oracle: from 0.609 at single seed to 0.743 $[0.725, 0.762]$ at $K=5$ best-of, a $+0.134$ absolute within-Envisage gain. Although this oracle ceiling sits above the GT-paste-no-composite control (0.703, Table 19), that comparison is structural rather than substantive: $K=5$ Envisage has $E \approx 1.0$ (the hard-mask composite preserves the input outside the surgical mask) while GT-paste has $E = 0$; D (FIQA realism) is also higher on a clean FLUX composite than on a clinical postoperative photograph by ≈ 0.15 . The relevant baseline for the $K=5$ oracle is $K=1$, not GT-paste; we report it as a within-pipeline candidate-space ceiling, not as a claim of superiority over the postoperative target. The open problem is again the design of a non-oracle candidate ranker that can recover the $K=5$ oracle gain at deployment time. Per-backbone $K=5$ best-of comparisons are reported in Table 23; Envisage produces the highest $K=5$ oracle ceiling among tested backbones.

Table 22: Per-case best-of- $K=5$ SurgicalScore oracle on $N=207$ (Envisage cases with all five seeds scored). Bootstrap 95% CI ($B=10,000$). The $K=5$ best-of mean of 0.743 exceeds both the strongest single seed (0.609, seed 42) and the GT-paste-no-composite calibration anchor (0.703). Source: `evaluation/strict_n211_baselines/surgicalscores_v5_envisage_seed*_n211.json`.

Strategy	N	Mean SS	95% CI
Single seed 42 (headline)	207	0.609	$[0.584, 0.634]$
Mean over 5 seeds (ensemble)	207	0.594	$[0.576, 0.612]$
Best of 5 (oracle)	207	0.743	$[0.725, 0.762]$
GT-paste no-composite (calibration anchor)	211	0.703	$[0.649, 0.756]$

U Backbone Substitution Ablation

We substitute the FLUX.1-Fill-dev backbone in the Envisage pipeline (mask, preset prompt, hard-mask composite) with three alternative diffusion backbones, holding all other pipeline components fixed. The variant labelled *no_cn* retains FLUX.1-Fill-dev but drops the depth ControlNet conditioning entirely; *icedit_lora* loads the ICEdit-MoE-LoRA [Zhang et al., 2025] on FLUX.1-Fill-dev as an off-the-shelf editing adapter; *kontext* replaces the inpainting backbone with FLUX.1-Kontext-dev [Batifol et al., 2025], a text-conditioned image-to-image model without native mask conditioning (we apply the Envisage hard-mask composite post-hoc); *sdxl_inpaint* replaces with the public Stable Diffusion XL inpainting checkpoint [Podell et al., 2024].

Three readings of these backbone-swap results are consistent with the data. First, FLUX.1-Fill-dev is a mask-aware inpainting backbone whose native boundary handling already approximates the

Table 23: Backbone substitution on $N=211$, mean SurgicalScore [95% CI] from per-case bootstrap ($B=10,000$) at $K=1$ (seed 42 only) and $K=5$ best-of (per-case oracle selection across seeds {42, 123, 456, 789, 1024}). At $K=1$, no_cn matches Envisage within CI; at $K=5$ best-of, Envisage edges ahead by +0.026 (0.743 vs 0.717), indicating that depth ControlNet conditioning produces candidate-space diversity that an oracle ranker can exploit even when the seed-mean is unchanged. Both FLUX.1-Fill family backbones substantially exceed SDXL-Inpaint, ICEdit-MoE-LoRA, and FLUX.1-Kontext-dev; Kontext underperforms because it lacks native inpainting and the post-hoc composite cannot recover mask-region structure that the img2img backbone overwrote. Source: evaluation/strict_n211_baselines/surgicalscores_v5*_seed*_n211.json.

Backbone	N	K=1	K=1 95% CI	K=5 best-of
Envisage (FLUX.1-Fill-dev + depth ControlNet)	207	0.609	[0.584, 0.634]	0.743 [0.725, 0.761]
no_cn (FLUX.1-Fill-dev native, no CN)	210	0.611	[0.585, 0.635]	0.717 [0.695, 0.738]
sdxl_inpaint (SDXL-Inpaint)	209	0.571	[0.544, 0.599]	0.731 [0.711, 0.750]
icedit_lora (FLUX.1-Fill-dev + ICEdit-MoE-LoRA)	190	0.533	[0.504, 0.562]	0.686 [0.661, 0.710]
kontext (FLUX.1-Kontext-dev)	104	0.318	[0.253, 0.382]	0.458 [0.386, 0.527]

structural information the depth ControlNet provides at single-seed scale; the conditioning becomes discriminative only when multiple candidates are scored. Second, the same mechanism that makes the composite operation dominate full-image identity metrics also dampens any backbone-level effect on SurgicalScore; the metric mostly reads the composite, not the diffusion backbone. Third, the depth ControlNet’s contribution is candidate-space-diversity rather than seed-mean: no_cn matches Envisage at $K=1$ within CI but trails by +0.026 at $K=5$ best-of, suggesting ControlNet expands the set of plausible per-seed outputs in a way that an oracle ranker can exploit. We retain the depth ControlNet in Envisage’s reference configuration to preserve qualitative profile-shape control observed in the surgeon review (Section 5.1); a fully ablated variant is reported here for transparency.

V Feathering Radius Robustness

We sweep the Gaussian-feathering radius σ used in the GT-paste control across $\sigma \in \{0, 4, 8, 12, 16, 24, 50\}$ pixels and re-score under the standard SurgicalScore protocol on $N=211$. SurgicalScore is essentially flat across this range:

Table 24: GT-paste control SurgicalScore as a function of feathering radius σ on $N=211$. Bootstrap CIs nearly identical; the metric is robust to feathering choice, which validates the production $\sigma=12$ –15 setting. The fact that even $\sigma=50$ (heavy spatial blending) does not collapse SS confirms that the dominant constraint on A and B is global landmark prediction on the composite, not local seam visibility. Source: evaluation/strict_n211_baselines/surgicalscores_v5_oracle_sigma*_n211.json.

σ (px)	N	Mean SS	95% CI
0 (hard mask)	207	0.543	[0.507, 0.578]
4	207	0.547	[0.511, 0.582]
8	207	0.547	[0.511, 0.582]
12 (production)	207	0.546	[0.510, 0.582]
16	208	0.546	[0.509, 0.582]
24	208	0.551	[0.515, 0.586]
50	209	0.556	[0.522, 0.589]

W Component Correlation and Discriminative Power

We also assessed discriminative power: for each component, we flagged cases falling below the 25th percentile and counted those flagged *only* by that component. Each of the five components uniquely flags 25–42% of its threshold-flagged cases (A: 17/53; B: 22/53; C: 13/53; D: 15/53; E: 15/53).

Table 25: Pearson correlation matrix of SurgicalScore components A–E across the $N=211$ Envisage cohort. Off-diagonal magnitudes are uniformly below 0.17, consistent with low linear redundancy among components. Source: `evaluation/strict_n211_baselines/correlation_v1.json`.

	A	B	C	D	E
A	+1.000	−0.014	+0.166	+0.034	+0.088
B	−0.014	+1.000	+0.096	−0.103	−0.084
C	+0.166	+0.096	+1.000	+0.066	+0.146
D	+0.034	−0.103	+0.066	+1.000	+0.022
E	+0.088	−0.084	+0.146	+0.022	+1.000

No component is redundant: each captures failures the others miss. Combined with the low pairwise correlation, this supports treating A–E as independent signals rather than a single underlying factor.

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Justification: The contributions are (1) the methodological diagnosis that full-image identity metrics are structurally confounded under hard-composited edits, with outside-mask SSIM above 0.999 by construction and replicating across blepharoplasty, rhinoplasty, and rhytidectomy; (2) the SurgicalScore benchmark protocol (five components A–E with weighted composite) and matched HDA paired splits ($N=27/21/9$) released as infrastructure; (3) cross-method, cross-procedure characterization on the $N=211$ rhinoplasty cohort showing no method achieves positive identity-space gain over the unedited input proxy, with Envisage producing the smallest residual gap (-0.048 , paired permutation $p < 10^{-4}$ against each baseline); (4) Envisage as a depth-conditioned FLUX-Fill reference pipeline serving as a pre-consult goal-visualization tool, not a patient-specific outcome predictor. All four are directly traceable to evaluation files (`evaluation/baselines/strict_n211_all_methods.json`, `strict_n211_baselines/`, `cross_proc_baselines/`, `cohort_expansion_2026-05-04/`). No claim exceeds what the referenced data support.

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 5.1 contains a Limitations paragraph naming the headline scope constraint (Envisage is a pre-consult goal-visualization tool, not a patient-specific outcome predictor); detailed limitations are in Appendix B, including: inside-mask accuracy does not exceed the pre-operative input proxy for any evaluated method; the matched test split is small ($N=21$); single-seed baselines introduce asymmetric variance; 2D geometry cannot represent three-dimensional rhinoplasty reshaping; gate thresholds are heuristic and not clinically validated; the FLUX.1-dev license is non-commercial; no cosmetic panel rating study has been conducted; and Monk Skin Tone coverage is preliminary with five of ten tones absent.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: No formal theoretical results are claimed. The relevant guarantee is the directly measured outside-mask SSIM exceeding 0.999 across all evaluated cohorts, a property of the hard-mask composite $C = M \odot G + (1 - M) \odot I$. An empirical Lipschitz characterization of ArcFace under masked perturbations ($L_{p95} \leq 1.2 \times 10^{-5}$ on 51 cases $\times$ 5 noise magnitudes = 255 perturbation samples; Appendix M) is reported as a numerical bound, not as a theorem.

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4. Experimental result reproducibility

Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

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Justification: Section 4 specifies model names (FLUX.1-dev, FLUX.1-Fill-dev, FLUX.1-Kontext-dev, Depth Anything V2 Small, InsightFace buffalo_l), inference hyperparameters (guidance 3.5, strength 0.75, ControlNet scale 0.5, 20 steps, BF16, 512 $\times$ 512), hardware (single NVIDIA L40S, 48 GB), inference time (20 s per image), and random seeds (seed 42 for splits and bootstrap). DPO training hyperparameters are in Appendix H. The released code package contains all evaluation harnesses.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to reproduce the main results?

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Justification: The SurgicalScore evaluation package, the Envisage pipeline code, and the HDA matched test split manifests/indices ($N=27$ blepharoplasty / $N=21$ rhinoplasty / $N=9$ rhytidectomy case identifiers and per-case metric outputs) are released at the supplementary material (zipped code bundle submitted alongside this paper PDF). The HDA *images* are not redistributed; they are obtained from the original authors [Rathgeb et al., 2020] under their research-use terms. The supplementary repository contains evaluation scripts, preset configurations, split manifests, and a README with step-by-step reproduction instructions.

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Question: Does the paper report error bars suitably and correctly defined?

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Justification: The headline $N=211$ ArcFace results in Table 1 include 95% percentile bootstrap CIs ($B=10,000$) over per-case scores. Table 2 reports paired sign-flip permutation tests against each baseline on the four-way detected intersection ($N=205$, $p < 10^{-4}$ for ICEdit, FLUX.1-Kontext-dev, and InstructPix2Pix). The $N=211$ SurgicalScore comparison (Tables 14, 15) carries per-case bootstrap CIs and paired permutation p -values on the four-way detected intersection ($N=188$). Mask-crop LPIPS (Table 16) and component correlation (Table 25) carry bootstrap CIs and discriminative-power statistics. Sensitivity sweeps (Tables 11, 12, 13) defend the heuristic thresholds. Single-seed legacy baselines (Table 5) do not have per-case bootstrap; the $\dagger$ notation distinguishes seed-level from per-case bootstrap.

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Justification: Section 4 states single NVIDIA L40S (48 GB), BF16 precision, 20 seconds per image inference. Appendix H discloses that DPO training ran on a single L40S for 2000 steps. The full matched-pool evaluation ($N=21$, 5 seeds) required approximately 35 GPU-minutes at this rate.

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893 code repository contains only evaluation harnesses and preset configurations. Architectural
894 safeguards: the hard-mask composite restricts generation to the surgical region; the identity
895 gate (≥ 0.65 ArcFace) rejects outputs with identity drift; the deterministic TPS fallback
896 (M5) activates when all diffusion candidates fail. The paper includes a prominent disclaimer
897 that outputs are simulation only and not for clinical use, and notes the non-commercial
898 FLUX.1-dev license.

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905 under its published research-use terms [Rathgeb et al., 2020]; InsightFace buffalo_1 is used
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915 bundle submitted alongside this paper PDF): (1) the SurgicalScore evaluation package with
916 documented API, usage examples, and unit tests; (2) the Envisage reference pipeline with a
917 README and step-by-step reproduction guide; (3) HDA matched test split manifests/indices
918 for $N=27$ blepharoplasty, $N=21$ rhinoplasty, and $N=9$ rhytidectomy paired cases (case
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930 The surgeon review described in Section 5 is an informal visual check by one expert reviewer
931 of generated output quality on six outputs selected to span SurgicalScore range and visible
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